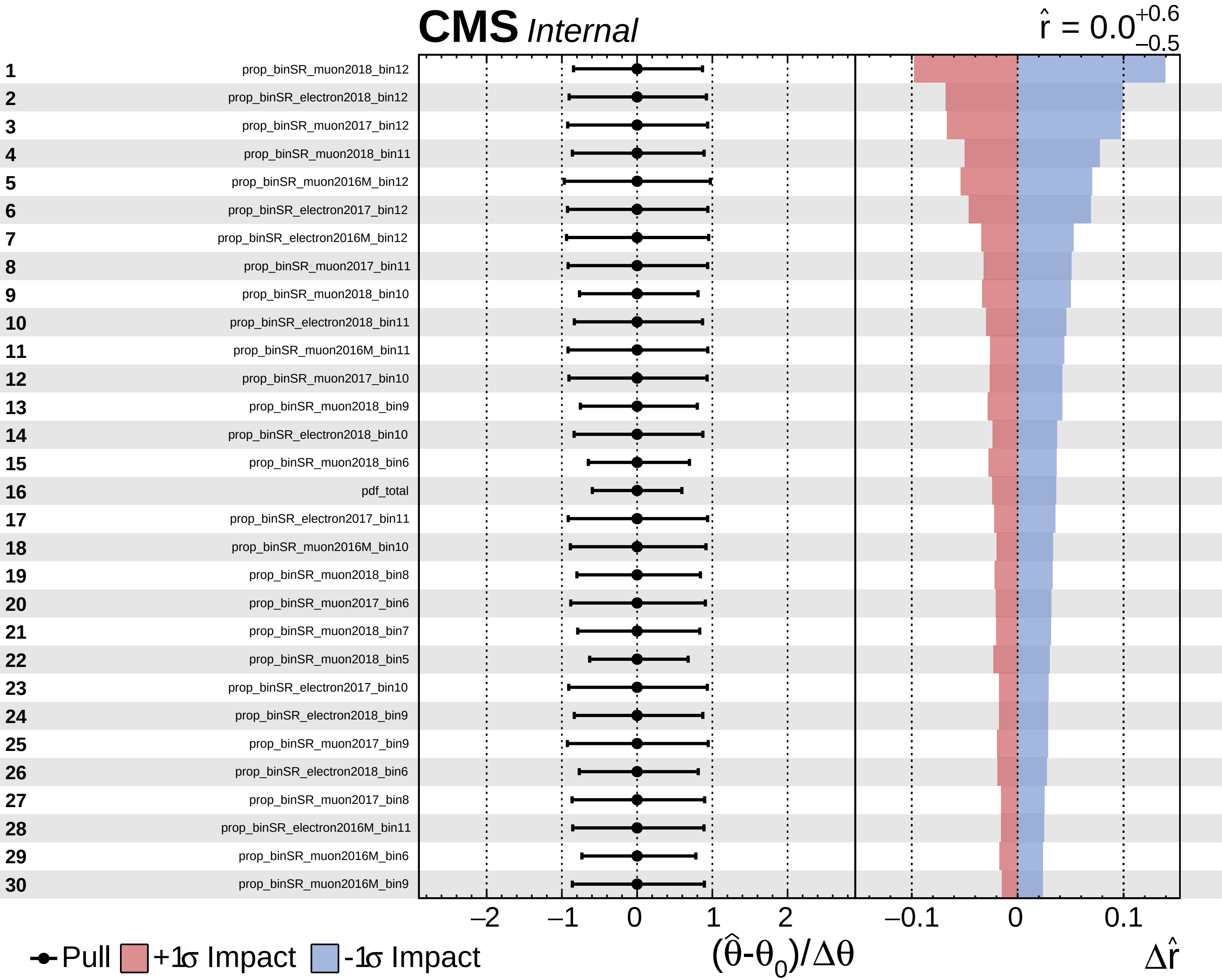
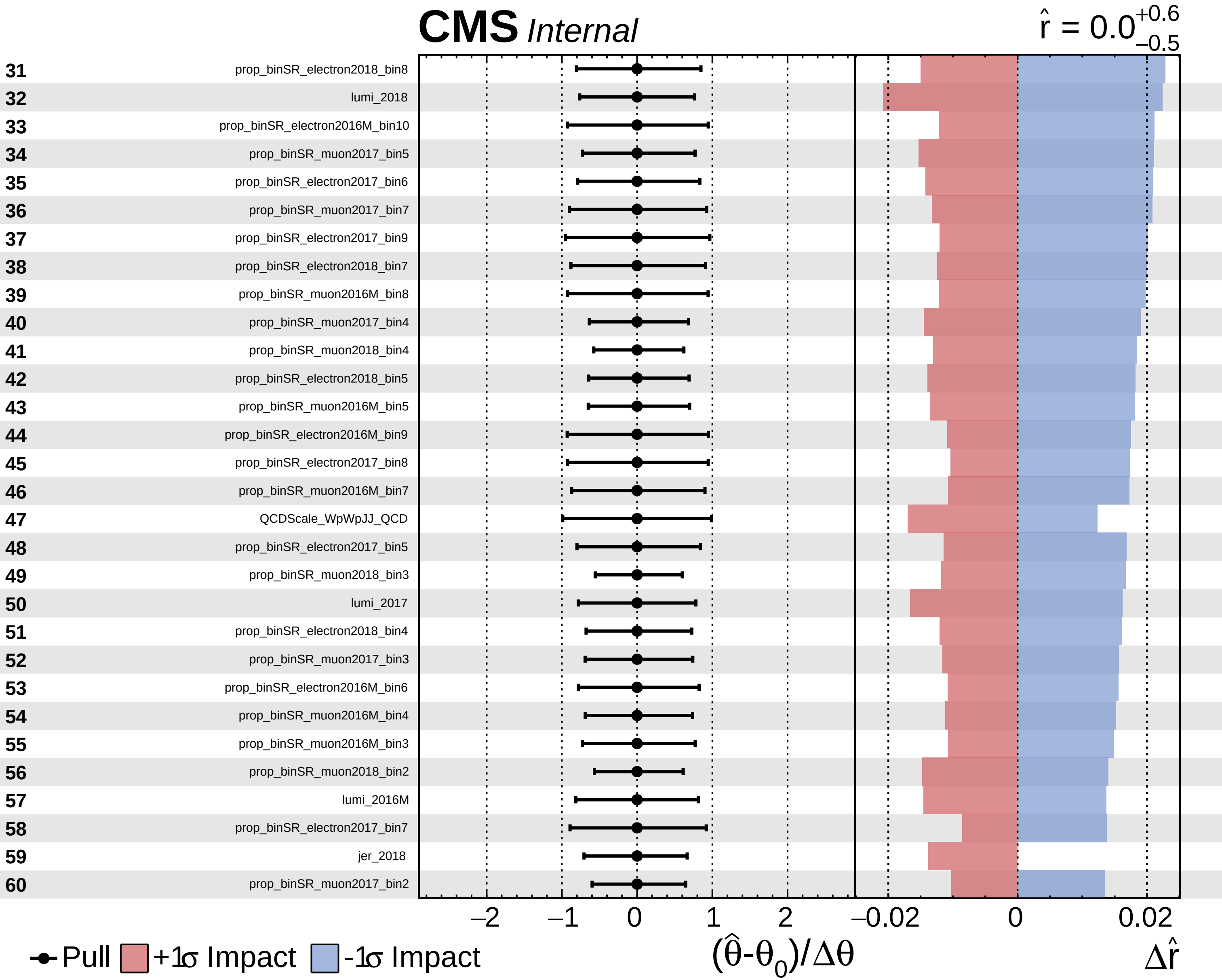
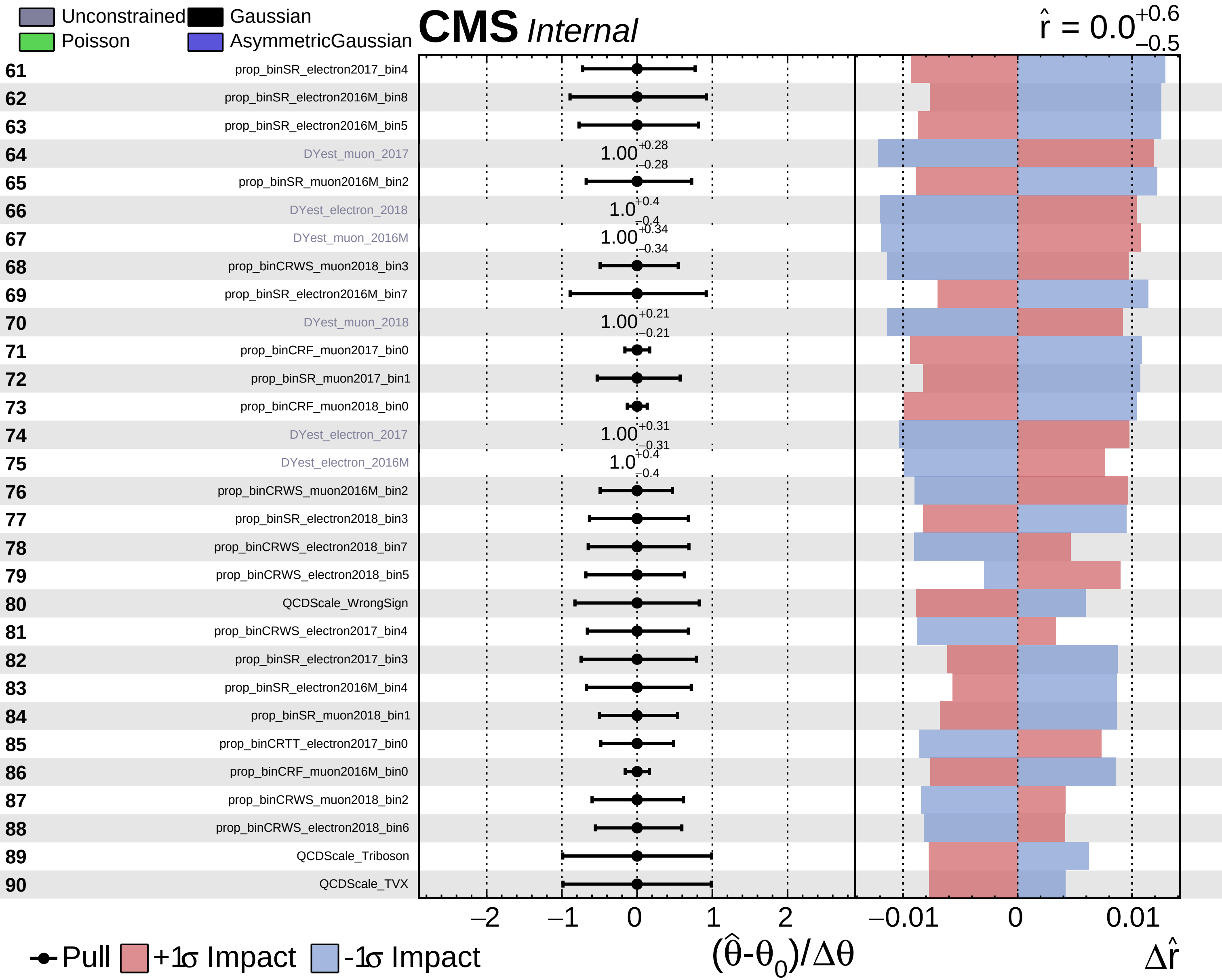
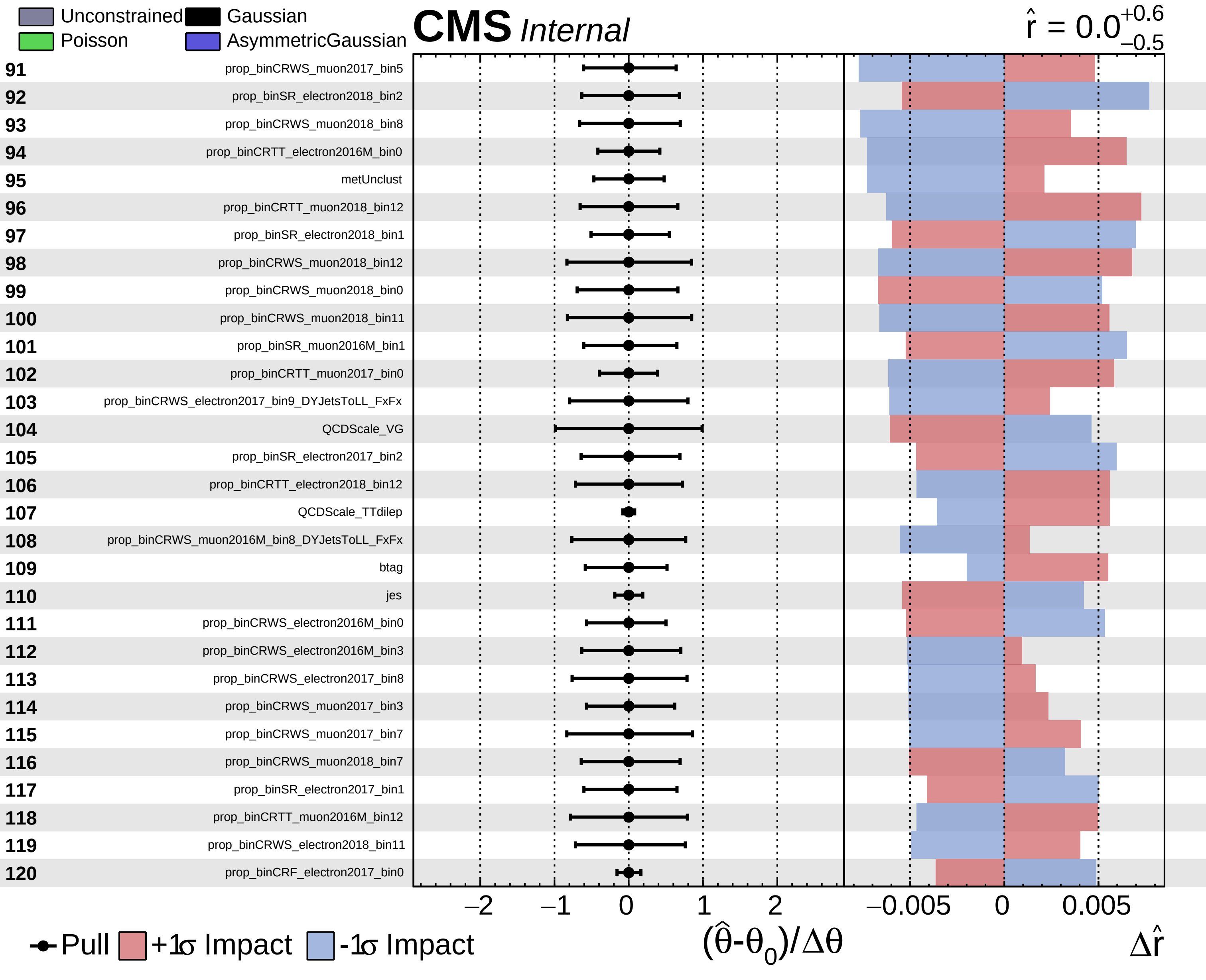

CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
1 prop_binSR_muon2018_bin12
2 prop_binSR_electron2018_bin12
3 prop_binSR_muon2017_bin12
4 prop_binSR_muon2018_bin11
5 prop_binSR_muon2016M_bin12
6 prop_binSR_electron2017_bin12
7 prop_binSR_electron2016M_bin12
8 prop_binSR_muon2017_bin11
9 prop_binSR_muon2018_bin10
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2016M_bin11
12 prop_binSR_muon2017_bin10
13 prop_binSR_muon2018_bin9
14 prop_binSR_electron2018_bin10
15 prop_binSR_muon2018_bin6
16 pdf_total
17 prop_binSR_electron2017_bin11
18 prop_binSR_muon2016M_bin10
19 prop_binSR_muon2018_bin8
20 prop_binSR_muon2017_bin6
21 prop_binSR_muon2018_bin7
22 prop_binSR_muon2018_bin5
23 prop_binSR_electron2017_bin10
24 prop_binSR_electron2018_bin9
25 prop_binSR_muon2017_bin9
26 prop_binSR_electron2018_bin6
27 prop_binSR_muon2017_bin8
28 prop_binSR_electron2016M_bin11
29 prop_binSR_muon2016M_bin6
30 prop_binSR_muon2016M_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

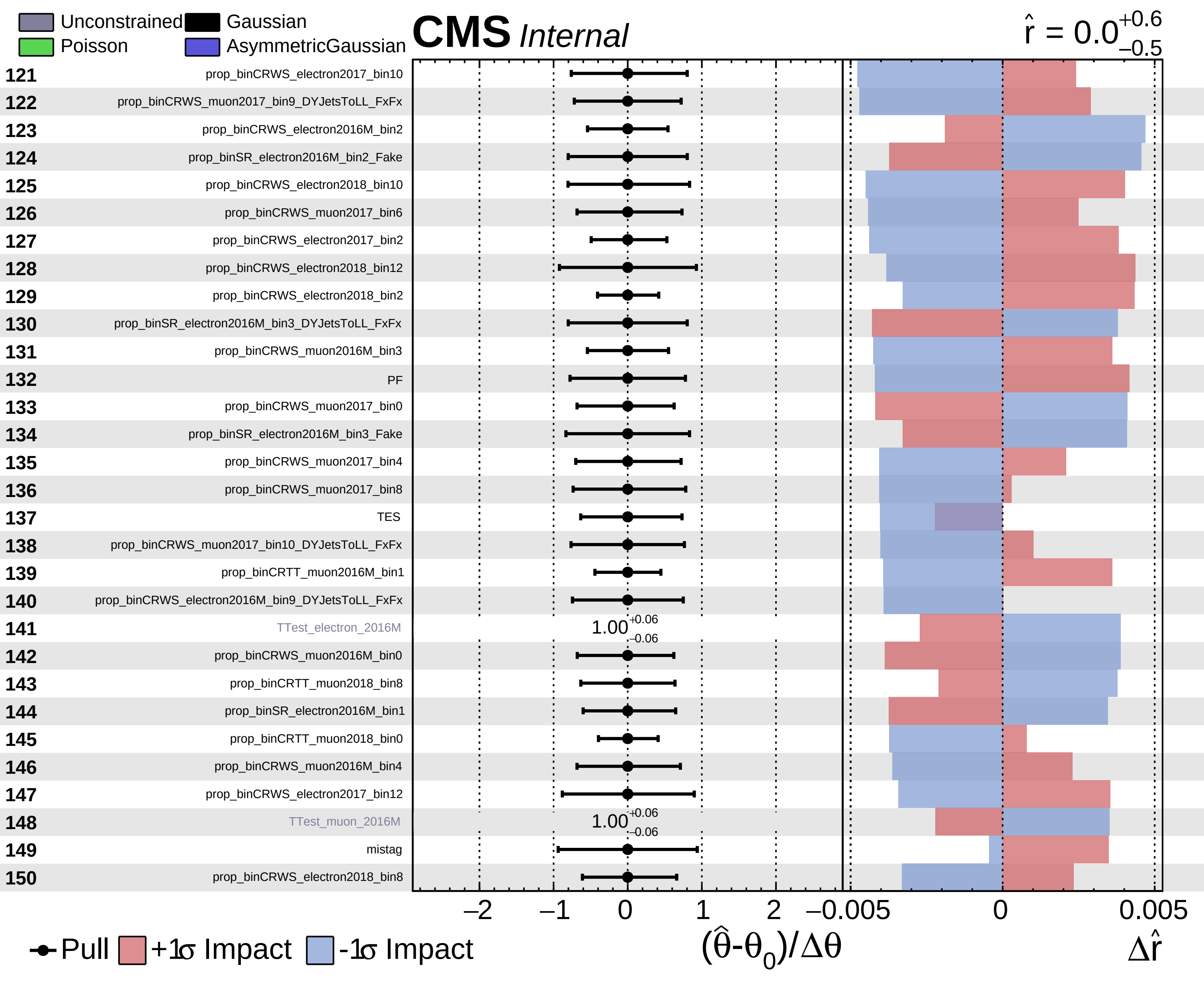

CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
31 prop_binSR_electron2018_bin8
32 lumi_2018
33 prop_binSR_electron2016M_bin10
34 prop_binSR_muon2017_bin5
35 prop_binSR_electron2017_bin6
36 prop_binSR_muon2017_bin7
37 prop_binSR_electron2017_bin9
38 prop_binSR_electron2018_bin7
39 prop_binSR_muon2016M_bin8
40 prop_binSR_muon2017_bin4
41 prop_binSR_muon2018_bin4
42 prop_binSR_electron2018_bin5
43 prop_binSR_muon2016M_bin5
44 prop_binSR_electron2016M_bin9
45 prop_binSR_electron2017_bin8
46 prop_binSR_muon2016M_bin7
47 QCDScale_WpWpJJ_QCD
48 prop_binSR_electron2017_bin5
49 prop_binSR_muon2018_bin3
50 lumi_2017
51 prop_binSR_electron2018_bin4
52 prop_binSR_muon2017_bin3
53 prop_binSR_electron2016M_bin6
54 prop_binSR_muon2016M_bin4
55 prop_binSR_muon2016M_bin3
56 prop_binSR_muon2018_bin2
57 lumi_2016M
58 prop_binSR_electron2017_bin7
59 jer_2018
60 prop_binSR_muon2017_bin2
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

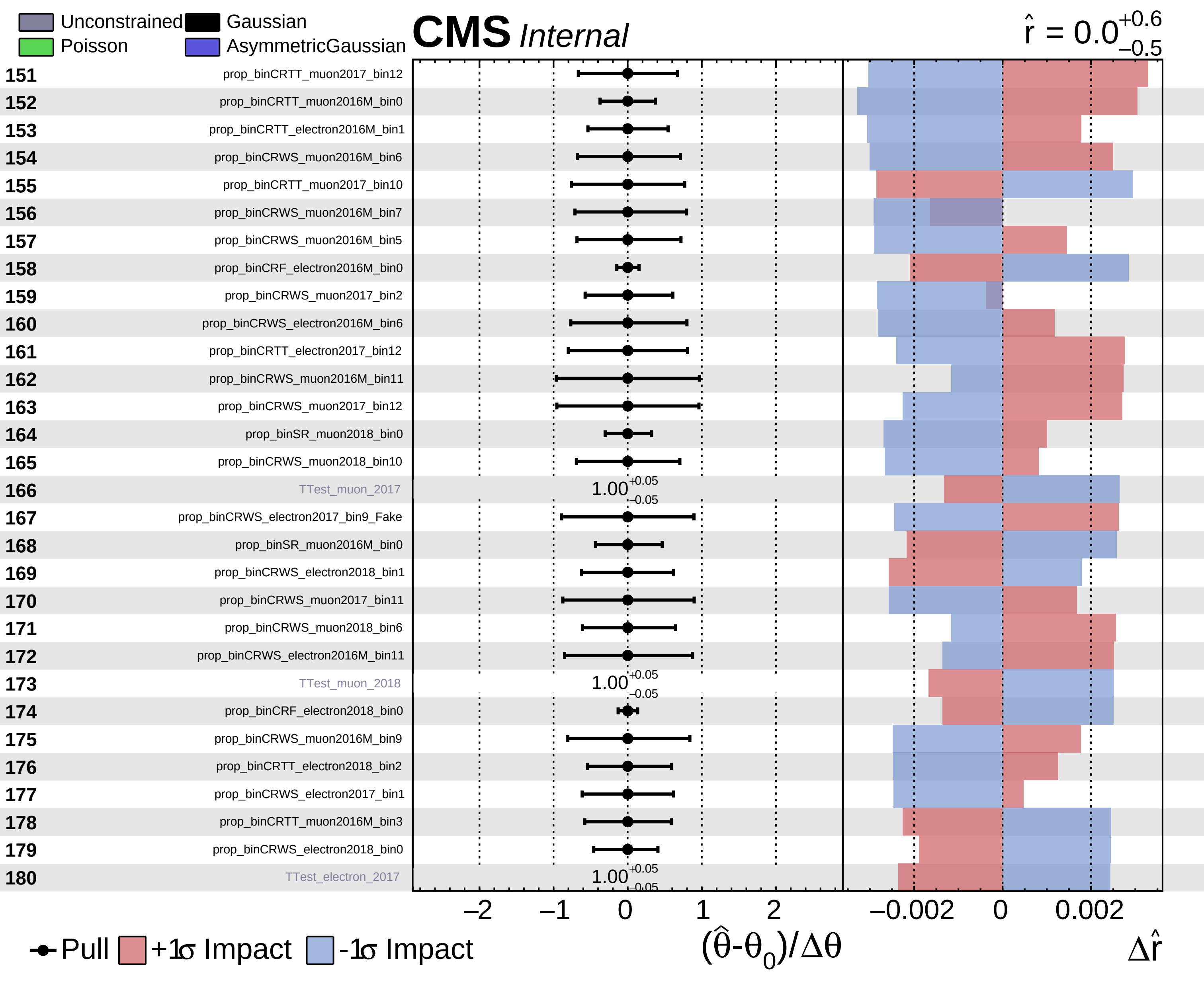


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
61 prop_binSR_electron2017_bin4
62 prop_binSR_electron2016M_bin8
63 prop_binSR_electron2016M_bin5
64 DYest_muon_2017 $1.00^{+0.28}_{-0.28}$
65 prop_binSR_muon2016M_bin2
66 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
67 DYest_muon_2016M $1.00^{+0.34}_{-0.34}$
68 prop_binCRWS_muon2018_bin3
69 prop_binSR_electron2016M_bin7
70 DYest_muon_2018 $1.00^{+0.21}_{-0.21}$
71 prop_binCRF_muon2017_bin0
72 prop_binSR_muon2017_bin1
73 prop_binCRF_muon2018_bin0
74 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
75 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
76 prop_binCRWS_muon2016M_bin2
77 prop_binSR_electron2018_bin3
78 prop_binCRWS_electron2018_bin7
79 prop_binCRWS_electron2018_bin5
80 QCDScale_WrongSign
81 prop_binCRWS_electron2017_bin4
82 prop_binSR_electron2017_bin3
83 prop_binSR_electron2016M_bin4
84 prop_binSR_muon2018_bin1
85 prop_binCRTT_electron2017_bin0
86 prop_binCRF_muon2016M_bin0
87 prop_binCRWS_muon2018_bin2
88 prop_binCRWS_electron2018_bin6
89 QCDScale_Triboson
90 QCDScale_TVX
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

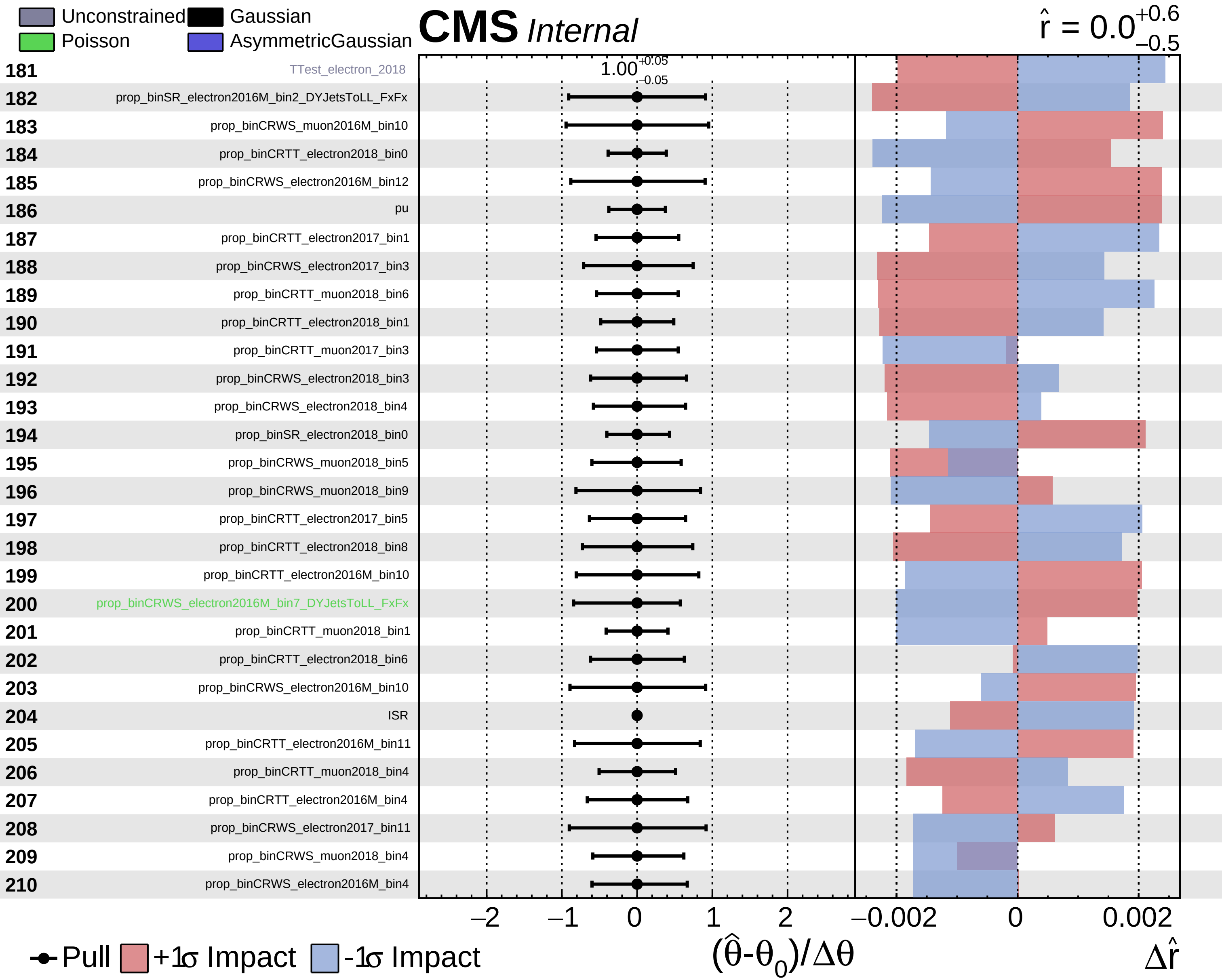


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
91 prop_binCRWS_muon2017_bin5
92 prop_binSR_electron2018_bin2
93 prop_binCRWS_muon2018_bin8
94 prop_binCRTT_electron2016M_bin0
95 metUnclust
96 prop_binCRTT_muon2018_bin12
97 prop_binSR_electron2018_bin1
98 prop_binCRWS_muon2018_bin12
99 prop_binCRWS_muon2018_bin0
100 prop_binCRWS_muon2018_bin11
101 prop_binSR_muon2016M_bin1
102 prop_binCRTT_muon2017_bin0
103 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
104 QCDScale_VG
105 prop_binSR_electron2017_bin2
106 prop_binCRTT_electron2018_bin12
107 QCDScale_TTdilep
108 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
109 btag
110 jes
111 prop_binCRWS_electron2016M_bin0
112 prop_binCRWS_electron2016M_bin3
113 prop_binCRWS_electron2017_bin8
114 prop_binCRWS_muon2017_bin3
115 prop_binCRWS_muon2017_bin7
116 prop_binCRWS_muon2018_bin7
117 prop_binSR_electron2017_bin1
118 prop_binCRTT_muon2016M_bin12
119 prop_binCRWS_electron2018_bin11
120 prop_binCRF_electron2017_bin0
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

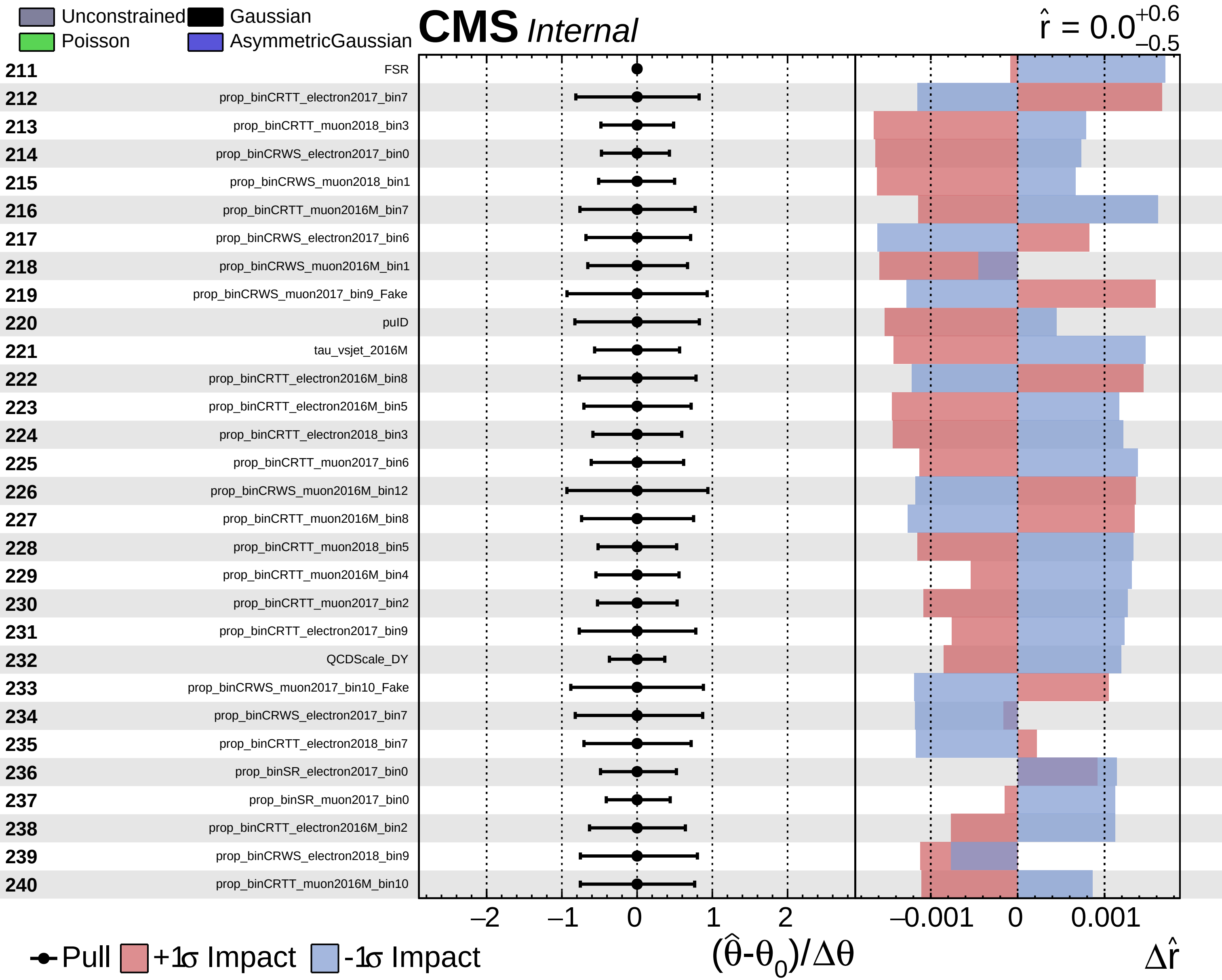

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
121 prop_binCRWS_electron2017_bin10
122 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
123 prop_binCRWS_electron2016M_bin2
124 prop_binSR_electron2016M_bin2_Fake
125 prop_binCRWS_electron2018_bin10
126 prop_binCRWS_muon2017_bin6
127 prop_binCRWS_electron2017_bin2
128 prop_binCRWS_electron2018_bin12
129 prop_binCRWS_electron2018_bin2
130 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
131 prop_binCRWS_muon2016M_bin3
132 PF
133 prop_binCRWS_muon2017_bin0
134 prop_binSR_electron2016M_bin3_Fake
135 prop_binCRWS_muon2017_bin4
136 prop_binCRWS_muon2017_bin8
137 TES
138 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
139 prop_binCRTT_muon2016M_bin1
140 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
141 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
142 prop_binCRWS_muon2016M_bin0
143 prop_binCRTT_muon2018_bin8
144 prop_binSR_electron2016M_bin1
145 prop_binCRTT_muon2018_bin0
146 prop_binCRWS_muon2016M_bin4
147 prop_binCRWS_electron2017_bin12
148 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
149 mistag
150 prop_binCRWS_electron2018_bin8
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



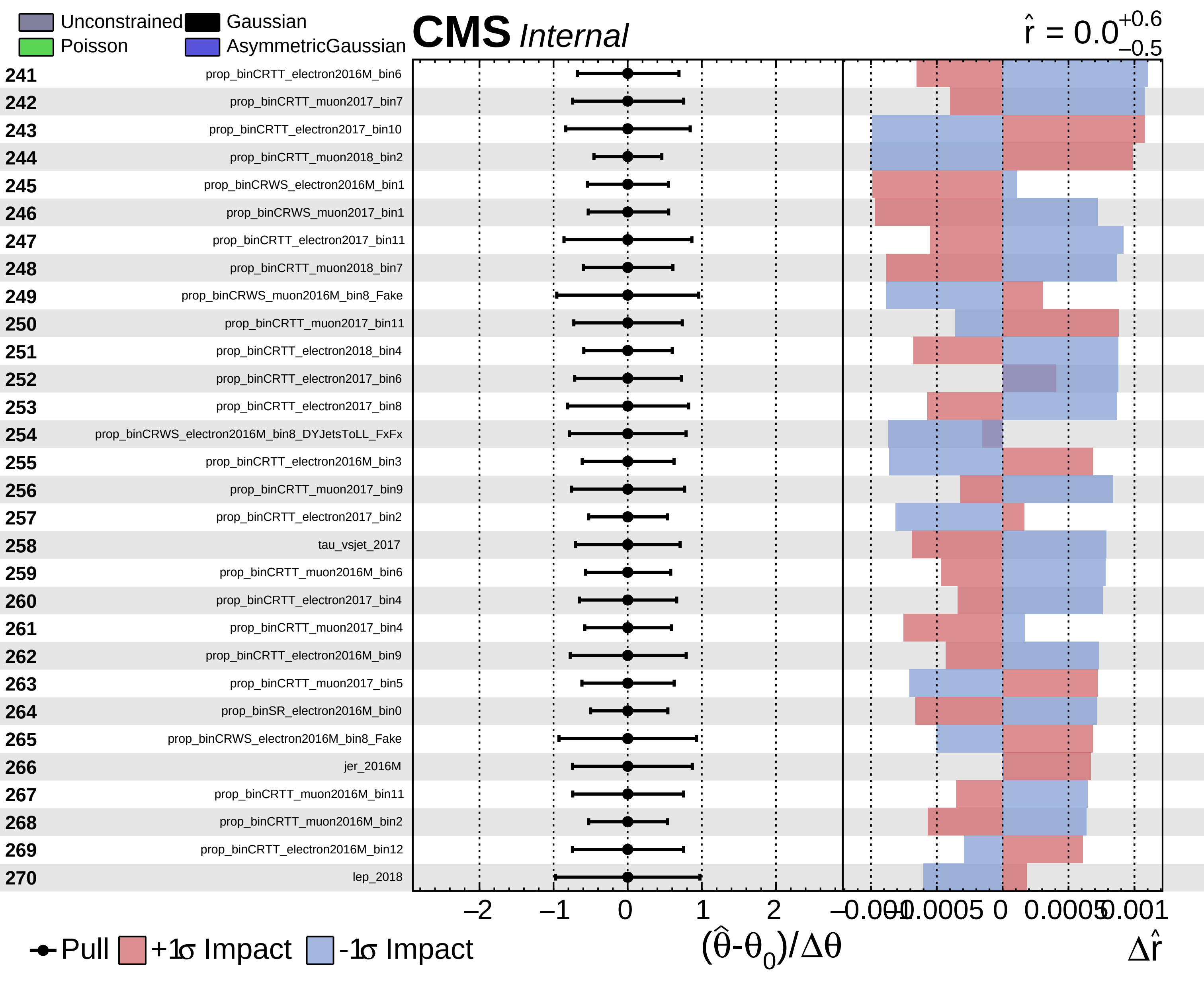
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
151 prop_binCRTT_muon2017_bin12
152 prop_binCRTT_muon2016M_bin0
153 prop_binCRTT_electron2016M_bin1
154 prop_binCRWS_muon2016M_bin6
155 prop_binCRTT_muon2017_bin10
156 prop_binCRWS_muon2016M_bin7
157 prop_binCRWS_muon2016M_bin5
158 prop_binCRF_electron2016M_bin0
159 prop_binCRWS_muon2017_bin2
160 prop_binCRWS_electron2016M_bin6
161 prop_binCRTT_electron2017_bin12
162 prop_binCRWS_muon2016M_bin11
163 prop_binCRWS_muon2017_bin12
164 prop_binSR_muon2018_bin0
165 prop_binCRWS_muon2018_bin10
166 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
167 prop_binCRWS_electron2017_bin9_Fake
168 prop_binSR_muon2016M_bin0
169 prop_binCRWS_electron2018_bin1
170 prop_binCRWS_muon2017_bin11
171 prop_binCRWS_muon2018_bin6
172 prop_binCRWS_electron2016M_bin11
173 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
174 prop_binCRF_electron2018_bin0
175 prop_binCRWS_muon2016M_bin9
176 prop_binCRTT_electron2018_bin2
177 prop_binCRWS_electron2017_bin1
178 prop_binCRTT_muon2016M_bin3
179 prop_binCRWS_electron2018_bin0
180 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



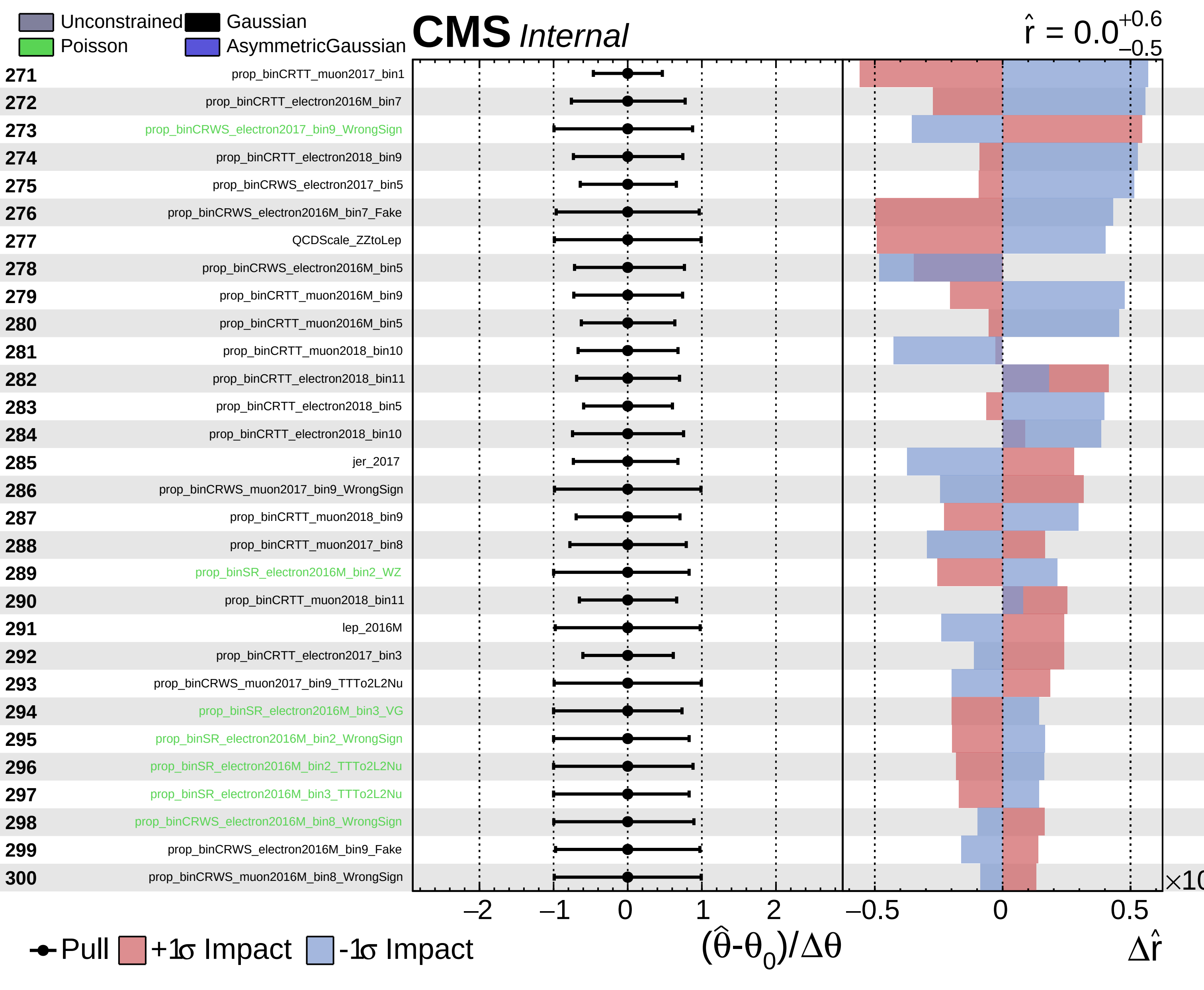
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
181 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
182 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
183 prop_binCRWS_muon2016M_bin10
184 prop_binCRTT_electron2018_bin0
185 prop_binCRWS_electron2016M_bin12
186 pu
187 prop_binCRTT_electron2017_bin1
188 prop_binCRWS_electron2017_bin3
189 prop_binCRTT_muon2018_bin6
190 prop_binCRTT_electron2018_bin1
191 prop_binCRTT_muon2017_bin3
192 prop_binCRWS_electron2018_bin3
193 prop_binCRWS_electron2018_bin4
194 prop_binSR_electron2018_bin0
195 prop_binCRWS_muon2018_bin5
196 prop_binCRWS_muon2018_bin9
197 prop_binCRTT_electron2017_bin5
198 prop_binCRTT_electron2018_bin8
199 prop_binCRTT_electron2016M_bin10
200 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
201 prop_binCRTT_muon2018_bin1
202 prop_binCRTT_electron2018_bin6
203 prop_binCRWS_electron2016M_bin10
204 ISR
205 prop_binCRTT_electron2016M_bin11
206 prop_binCRTT_muon2018_bin4
207 prop_binCRTT_electron2016M_bin4
208 prop_binCRWS_electron2017_bin11
209 prop_binCRWS_muon2018_bin4
210 prop_binCRWS_electron2016M_bin4
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



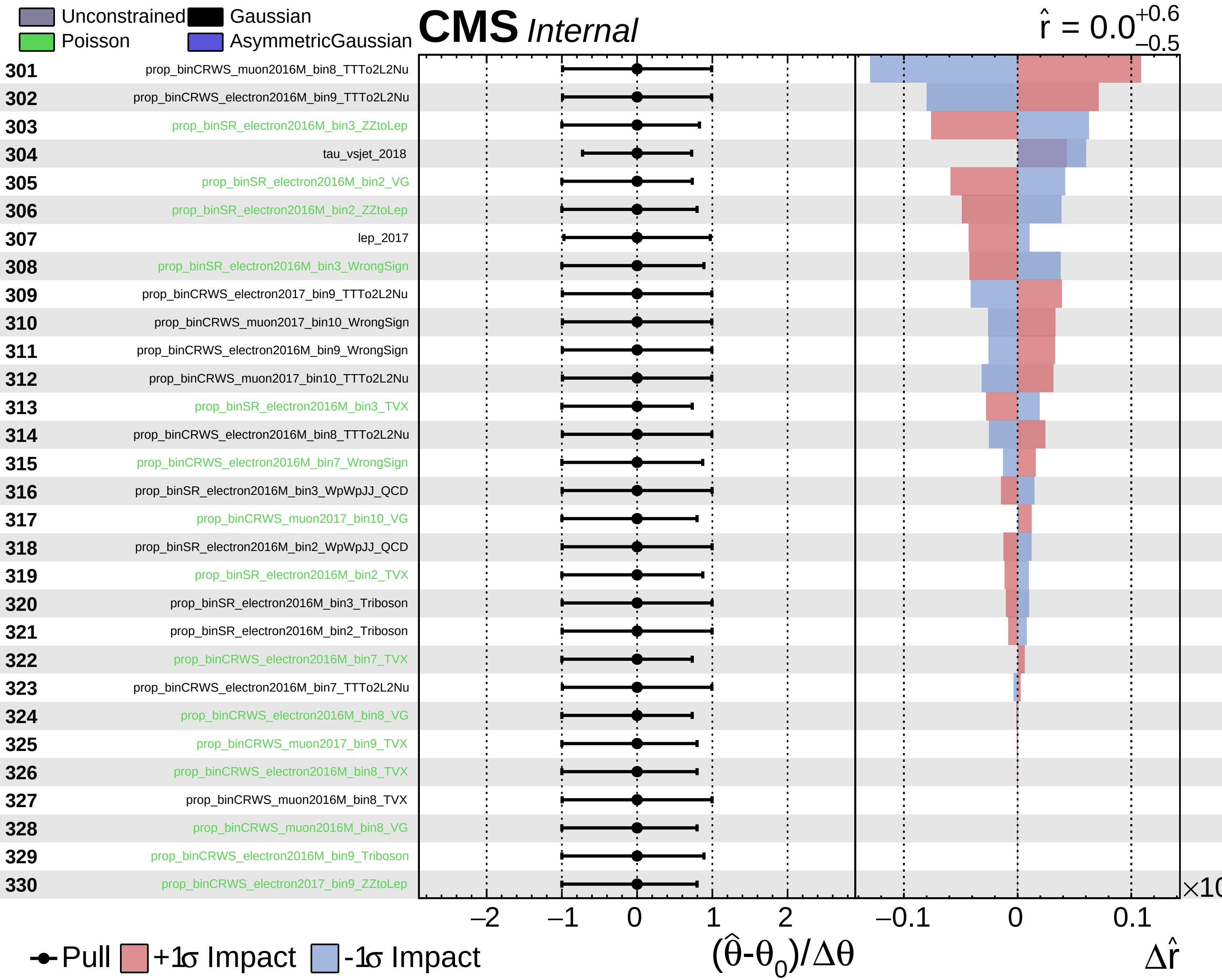
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
211 FSR
212 prop_binCRTT_electron2017_bin7
213 prop_binCRTT_muon2018_bin3
214 prop_binCRWS_electron2017_bin0
215 prop_binCRWS_muon2018_bin1
216 prop_binCRTT_muon2016M_bin7
217 prop_binCRWS_electron2017_bin6
218 prop_binCRWS_muon2016M_bin1
219 prop_binCRWS_muon2017_bin9_Fake
220 puID
221 tau_vsjet_2016M
222 prop_binCRTT_electron2016M_bin8
223 prop_binCRTT_electron2016M_bin5
224 prop_binCRTT_electron2018_bin3
225 prop_binCRTT_muon2017_bin6
226 prop_binCRWS_muon2016M_bin12
227 prop_binCRTT_muon2016M_bin8
228 prop_binCRTT_muon2018_bin5
229 prop_binCRTT_muon2016M_bin4
230 prop_binCRTT_muon2017_bin2
231 prop_binCRTT_electron2017_bin9
232 QCDScale_DY
233 prop_binCRWS_muon2017_bin10_Fake
234 prop_binCRWS_electron2017_bin7
235 prop_binCRTT_electron2018_bin7
236 prop_binSR_electron2017_bin0
237 prop_binSR_muon2017_bin0
238 prop_binCRTT_electron2016M_bin2
239 prop_binCRWS_electron2018_bin9
240 prop_binCRTT_muon2016M_bin10
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

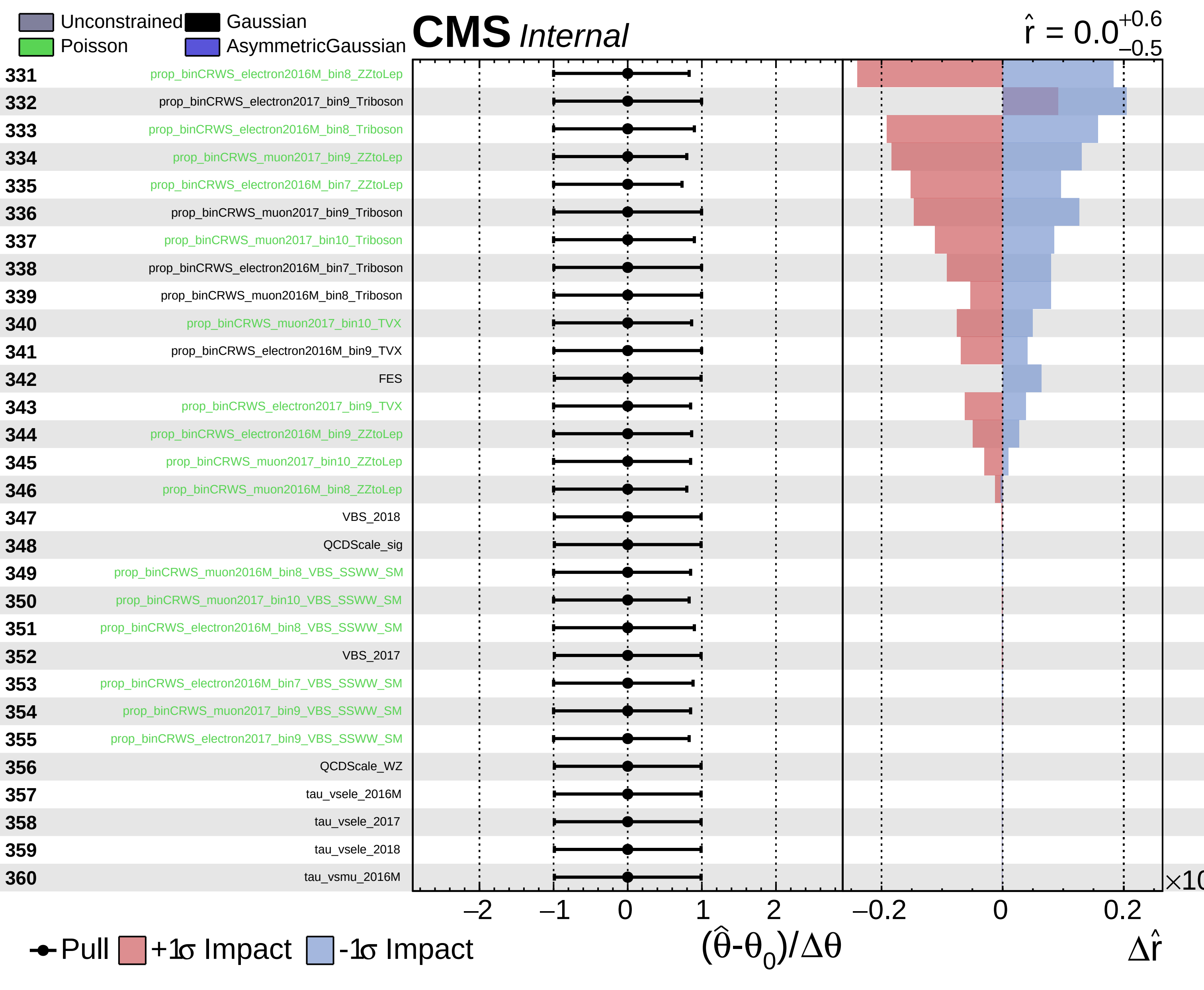


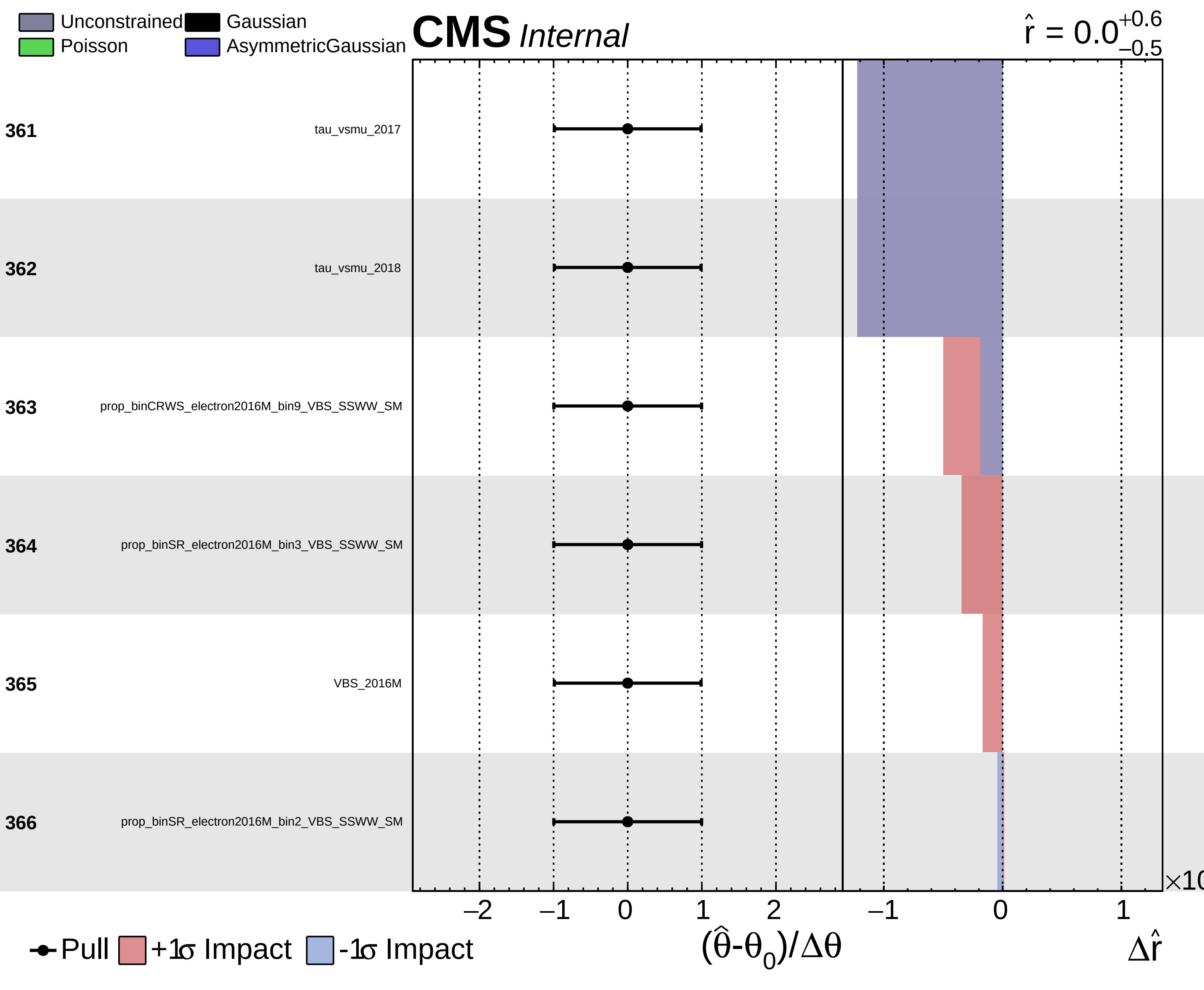
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
241 prop_binCRTT_electron2016M_bin6
242 prop_binCRTT_muon2017_bin7
243 prop_binCRTT_electron2017_bin10
244 prop_binCRTT_muon2018_bin2
245 prop_binCRWS_electron2016M_bin1
246 prop_binCRWS_muon2017_bin1
247 prop_binCRTT_electron2017_bin11
248 prop_binCRTT_muon2018_bin7
249 prop_binCRWS_muon2016M_bin8_Fake
250 prop_binCRTT_muon2017_bin11
251 prop_binCRTT_electron2018_bin4
252 prop_binCRTT_electron2017_bin6
253 prop_binCRTT_electron2017_bin8
254 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
255 prop_binCRTT_electron2016M_bin3
256 prop_binCRTT_muon2017_bin9
257 prop_binCRTT_electron2017_bin2
258 tau_vsjet_2017
259 prop_binCRTT_muon2016M_bin6
260 prop_binCRTT_electron2017_bin4
261 prop_binCRTT_muon2017_bin4
262 prop_binCRTT_electron2016M_bin9
263 prop_binCRTT_muon2017_bin5
264 prop_binSR_electron2016M_bin0
265 prop_binCRWS_electron2016M_bin8_Fake
266 jer_2016M
267 prop_binCRTT_muon2016M_bin11
268 prop_binCRTT_muon2016M_bin2
269 prop_binCRTT_electron2016M_bin12
270 lep_2018
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
271 prop_binCRTT_muon2017_bin1
272 prop_binCRTT_electron2016M_bin7
273 prop_binCRWS_electron2017_bin9_WrongSign
274 prop_binCRTT_electron2018_bin9
275 prop_binCRWS_electron2017_bin5
276 prop_binCRWS_electron2016M_bin7_Fake
277 QCDScale_ZZtoLep
278 prop_binCRWS_electron2016M_bin5
279 prop_binCRTT_muon2016M_bin9
280 prop_binCRTT_muon2016M_bin5
281 prop_binCRTT_muon2018_bin10
282 prop_binCRTT_electron2018_bin11
283 prop_binCRTT_electron2018_bin5
284 prop_binCRTT_electron2018_bin10
285 jer_2017
286 prop_binCRWS_muon2017_bin9_WrongSign
287 prop_binCRTT_muon2018_bin9
288 prop_binCRTT_muon2017_bin8
289 prop_binSR_electron2016M_bin2_WZ
290 prop_binCRTT_muon2018_bin11
291 lep_2016M
292 prop_binCRTT_electron2017_bin3
293 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
294 prop_binSR_electron2016M_bin3_VG
295 prop_binSR_electron2016M_bin2_WrongSign
296 prop_binSR_electron2016M_bin2_TTTo2L2Nu
297 prop_binSR_electron2016M_bin3_TTTo2L2Nu
298 prop_binCRWS_electron2016M_bin8_WrongSign
299 prop_binCRWS_electron2016M_bin9_Fake
300 prop_binCRWS_muon2016M_bin8_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
301 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
302 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
303 prop_binSR_electron2016M_bin3_ZZtoLep
304 tau_vsjet_2018
305 prop_binSR_electron2016M_bin2_VG
306 prop_binSR_electron2016M_bin2_ZZtoLep
307 lep_2017
308 prop_binSR_electron2016M_bin3_WrongSign
309 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
310 prop_binCRWS_muon2017_bin10_WrongSign
311 prop_binCRWS_electron2016M_bin9_WrongSign
312 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
313 prop_binSR_electron2016M_bin3_TVX
314 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
315 prop_binCRWS_electron2016M_bin7_WrongSign
316 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
317 prop_binCRWS_muon2017_bin10_VG
318 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
319 prop_binSR_electron2016M_bin2_TVX
320 prop_binSR_electron2016M_bin3_Triboson
321 prop_binSR_electron2016M_bin2_Triboson
322 prop_binCRWS_electron2016M_bin7_TVX
323 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
324 prop_binCRWS_electron2016M_bin8_VG
325 prop_binCRWS_muon2017_bin9_TVX
326 prop_binCRWS_electron2016M_bin8_TVX
327 prop_binCRWS_muon2016M_bin8_TVX
328 prop_binCRWS_muon2016M_bin8_VG
329 prop_binCRWS_electron2016M_bin9_Triboson
330 prop_binCRWS_electron2017_bin9_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
331 prop_binCRWS_electron2016M_bin8_ZZtoLep
332 prop_binCRWS_electron2017_bin9_Triboson
333 prop_binCRWS_electron2016M_bin8_Triboson
334 prop_binCRWS_muon2017_bin9_ZZtoLep
335 prop_binCRWS_electron2016M_bin7_ZZtoLep
336 prop_binCRWS_muon2017_bin9_Triboson
337 prop_binCRWS_muon2017_bin10_Triboson
338 prop_binCRWS_electron2016M_bin7_Triboson
339 prop_binCRWS_muon2016M_bin8_Triboson
340 prop_binCRWS_muon2017_bin10_TVX
341 prop_binCRWS_electron2016M_bin9_TVX
342 FES
343 prop_binCRWS_electron2017_bin9_TVX
344 prop_binCRWS_electron2016M_bin9_ZZtoLep
345 prop_binCRWS_muon2017_bin10_ZZtoLep
346 prop_binCRWS_muon2016M_bin8_ZZtoLep
347 VBS_2018
348 QCDScale_sig
349 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
350 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
351 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
352 VBS_2017
353 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
354 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
355 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
356 QCDScale_WZ
357 tau_vsele_2016M
358 tau_vsele_2017
359 tau_vsele_2018
360 tau_vsmu_2016M
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.6}_{-0.5}$
361
tau_vsmu_2017
362
tau_vsmu_2018
363
prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
364
prop_binSR_electron2016M_bin3_VBS_SSWW_SM
365
VBS_2016M
366
prop_binSR_electron2016M_bin2_VBS_SSWW_SM
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$